Completing the Square

Exam code: 9MA0

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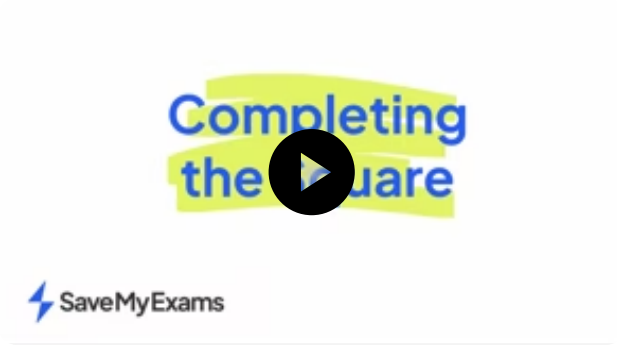

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Completing the square

What is completing the square?

- Completing the square is another method used to solve quadratic equations
- It simply means writing $y = ax^2 + bx + c$ in the form $y = a(x+p)^2 + q$
- It can be used to help find other information about the quadratic like coordinates of the turning point

How do I complete the square?

The method used will depend on the value of the coefficient of the $\mathbf{x^2}$ term in $y = ax^2 + bx + c$

When **a = 1**

- p is half of b
- q is c - p^2

e.g. $y = x^2 + 8x - 2$

STEP 1: "$p = \frac{b}{2}$" $p = \frac{8}{2} = 4$

STEP 2: "$q = c - p^2$" $q = -2 - 4^2 = -18$

STEP 3: WRITE THE ANSWER!
$y = (x+4)^2 - 18$

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Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- You first need to take a out as a factor of the x^2 and x terms
- Then continue as above

e.g. $y = 4x^2 + 16x + 5$

NO NEED TO INVOLVE "c"

STEP 1: FACTOR "a" ON RHS $y = 4[x^2 + 4x] + 5$

STEP 2: WORKING WITH $[x^2 + 4x]$ ONLY, "$p = \frac{b}{2}$"

$p = \frac{4}{2} = 2$

STEP 3: STILL WITH $[x^2 + 4x]$ ONLY, "$q = c - p^2$"

$q = 0 - 2^2 = -4$

STEP 4: EXPAND AND SIMPLIFY $y = 4[(x+2)^2 - 4] + 5$

$y = 4(x+2)^2 - 16 + 5$

$y = 4(x+2)^2 - 11$

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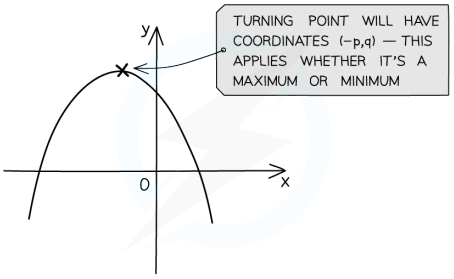
When is completing the square useful?

- Completing the square helps us find the **turning point** on a quadratic graph
- It can also help you **create the equation** of a quadratic when given the turning point



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- It can also be used to **prove** and/or **show** results using the fact that a squared term will always be greater than or equal to 0

e.g. $y = x^2 + 6x - 3$

$y = (x+3)^2 - 12$

$\therefore y \geqslant -12$

A SQUARED TERM WILL HAVE A MINIMUM VALUE OF 0

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- Sometimes the question will explicitly ask you to complete the square
- Sometimes it will even remind you of the form to write it in
- But sometimes it will expect you to spot that completing the square is what you need to do to help with other parts of the question... like finding turning points!

Worked Example

(a) Show $2x^2 + 6x + k > 0$ for $k > \frac{9}{2}$

(b) What can you say about the roots of the equation

$2x^2 + 6x + k = 0$ when $k = \frac{9}{2}$?

(a) ">0" IMPLIES A MINIMUM VALUE — SO COMPLETE THE SQUARE WITH "a≠1"

$2x^2 + 6x + k = 2[x^2 + 3x] + k$

STEP 1: FACTOR "a"

$= 2[(x + \frac{3}{2})^2 - \frac{9}{4}] + k$

STEP 2: "$p = \frac{b}{2}$"

STEP 3: "$q = c - p^2$"

NOTE WITH STEP 2 & STEP 3 YOU ARE ONLY WORKING WITHIN THE SQUARE [ ] BRACKETS

$= 2(x + \frac{3}{2})^2 - \frac{9}{2} + k$

$= 2(x + \frac{3}{2})^2 + k - \frac{9}{2}$

STEP 4: EXPAND AND SIMPLIFY

$2(x + \frac{3}{2})^2 \geq 0$ FOR ALL VALUES OF x

$\therefore 2(x + \frac{3}{2})^2 + k - \frac{9}{2} > 0$ WHEN $k - \frac{9}{2} > 0$

$k > \frac{9}{2}$

SHOW ALL STAGES IN A "SHOW THAT" QUESTION

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